EXP:2 - TRANSPORTATION PROBLEM

AIM:

To solve the Transportation Problem for minimizing the cost of transporting goods from multiple sources to multiple destinations using Python.

Procedure:

1. Install Required Libraries:

- Make sure you have numpy and scipy installed. You can install them using pip if not already installed.

  bash

  Copy code

  pip install numpy scipy

2. Define the Problem:

- Identify the cost matrix, supply array, and demand array.

3. Set Up the Problem in Python:

- Use the scipy.optimize.linprog function to set up and solve the Transportation Problem.

4. Run the Code:

- Execute the Python code to get the optimized result.

5. Analyze the Output:

- Interpret the results and validate the solution against the problem constraints.

CODE:

Code to solve a Linear Programming Problem using Python.

```
import numpy as np

from scipy.optimize import linprog


# Cost matrix (3 warehouses × 4 stores)

cost = np.array([

    [8, 6, 10, 9],

    [9, 7, 4, 2],

    [3, 4, 2, 5]
```

```
])

# Flatten cost matrix
c = cost.flatten()

# Supply and demand
supply = np.array([20, 30, 25])
demand = np.array([10, 25, 20, 20])

# Create equality constraints
A_eq = []
b_eq = []

# Supply constraints
for i in range(len(supply)):
    constraint = [0] * len(c)
    for j in range(len(demand)):
        constraint[i * len(demand) + j] = 1
    A_eq.append(constraint)
    b_eq.append(supply[i])

# Demand constraints
for j in range(len(demand)):
    constraint = [0] * len(c)
    for i in range(len(supply)):
        constraint[i * len(demand) + j] = 1
    A_eq.append(constraint)
    b_eq.append(demand[j])

# Solve using linprog
result = linprog(c, A_eq=A_eq, b_eq=b_eq, bounds=(0, None), method='highs')
```

```
# Reshape result
result_matrix = result.x.reshape(cost.shape)


# Output
print("Optimal cost:", result.fun)
print("Optimal transportation plan:")
print(result_matrix)
```

Output:

```
Optimal cost: 270.0
Optimal transportation plan:
[[ 0. 20.  0.  0.]
 [ 0.  0. 10. 20.]
 [10.  5. 10.  0.]]
```

Result:

The given transportation problem was solved successfully using Python, and the optimal transportation cost and shipment plan were obtained.